ASCIIgent: Agentic ASCII Art Generation

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Abstract

Visual abstraction is the process of discarding redundant low-level visual data—textures, photometrics, and uninformative pixel noise—and re-encoding semantic structures into a reduced, symbolic visual language. While recent advances in neural sketch synthesis like SketchAgent demonstrate that vision-language models (VLMs) can synthesize continuous vector strokes, we investigate whether this spatial capacity extends to a fundamentally more constrained and discrete medium: **ASCII art**. In ASCII art, a model must convey recognizable silhouettes using exactly 95 printable characters on a fixed 24×48 monospace grid. Characters here serve a dual role: simultaneously functioning as language tokens and spatial micro-primitives. This dichotomy introduces a core question: do autoregressive models draw through spatial reasoning or pure textual memorization, and does active visual perception enable meaningful self-correction? We introduce **ASCIIgent**, a training-free framework evaluating monolithic prompting baselines, agentic discrete tool-use loops, and multimodal visual feedback. Evaluating against structure-destroying scrambled controls to mitigate the text-reading bias of VLM judges, we reveal characteristic failure modes of monolithic generation (scanline drift, lexical leakage) and prove that intermediate visual feedback is essential for grounded spatial refinement, localized editing, and coherent keyframe animation.

1 Introduction and Related Work

Visual Abstraction and Agentic Sketching. Visual abstraction is a core capability of biological and computational visual perception [1]. Neural stroke models pioneered recurrent drawing representations [2], which later advanced toward semantic vector sketching [3]. Most recently, **SketchAgent** [4] formalized sketch generation as a continuous perception–reasoning–action loop: a frozen multimodal LLM iteratively emits Bézier stroke control points, inspects the updated rasterized canvas, and refines the drawing. Concurrently, **VideoSketcher** [5] demonstrated that dynamic sketching can be decomposed into an LLM-guided high-level structural plan coupled with diffusion priors. We adapt and generalize this agentic paradigm to discrete character media.

The ASCII Art Medium and its Challenges. ASCII art presents a uniquely rigid constraint: characters must occupy fixed coordinates in a monospace grid $\Sigma^{24\times48}$. Unlike continuous splines, characters possess discrete geometry: a forward slash `/` acts as a $45°$ line segment, an at-sign `@` acts as a dense tonal patch, and a period `.` functions as point stippling. Monolithic autoregressive generation suffers from **scanline drift**: generating left-to-right, top-to-bottom forces the model to commit to features at row 2 that must align with geometry at row 22 hundreds of tokens later without intermediate visual verification or backtracking. Public archives like `asciiart.eu` [6] document decades of human designs, which language models have internalized during pretraining.

The VLM Evaluation Confound. Evaluating ASCII art with multimodal models introduces a severe confounding factor: as shown by Wang et al. [7] and ASCIIBench [8], VLMs systematically suffer from text bias, recognizing concepts by reading characters rather than perceiving shapes. Our evaluation incorporates rigorous scrambled controls (*scramble-cells* and *scramble-ink*) to isolate genuine visual recognizability.

2 Methodology

2.1 Canvas Representation and Format Contract

The drawing canvas is defined as $C \in \Sigma^{H\times W}$ ($H = 24$, $W = 48$, $|\Sigma| = 95$). Monospace rasterization $\mathcal{R} : \Sigma^{H\times W} \to \mathbb{R}^{H'\times W'\times 3}$ renders glyphs at exact pixel offsets without font kerning. The system enforces strict parsing: canvas overflows or non-ASCII escapes are flagged in a `ParseReport` and treated as structural defects.

2.2 One-Shot Generation Baselines (P1)

We evaluate three controlled one-shot prompting strategies across seen and held-out categories:

1. **Zero-Shot:** Direct instruction: "Draw a {category} on the 24×48 canvas."
2. **Few-Shot:** Same prompt preceded by four structurally diverse exemplar categories (`fish`, `house`, `tree`, `sailboat`) strictly disjoint from test categories.
3. **Procedural (CoT):** The model explicitly plans the object's decomposition, bounding coordinates, and glyph selections in structured prose before emitting the grid.

2.3 Agentic Generation via Visual Feedback (P2)

Instead of monolithic emission, ASCIIgent interacts with an `AsciiCanvas` through a discrete spatial action vocabulary:

- `draw_line`$(r_1, c_1, r_2, c_2, \textit{char})$: Bresenham line interpolation with automatic directional glyph selection.
- `draw_rect`$(t, l, b, r, \textit{border}, \textit{fill})$: Axis-aligned bounding box.
- `shade_density`(t, l, b, r, ℓ): Tonal fill from an ordered density ramp: `' ' < '.' < ':' < '-' < '=' < '+' < '*' < '#' < '%' < '@'`.
- `patch_edit`$(r, c, \textit{lines})$: In-place localized subgrid overwrite.
- `erase_rect`(t, l, b, r): Region clearing.
- `finish`: Session termination.

Feedback Ablation. At the end of each turn, the canvas state is processed under three conditions: (1) **Visual (Ours):** rendered to a bitmap PNG and injected into the model's visual context; (2) **Text-Only:** presented as a raw character matrix; (3) **No-Feedback:** blind iteration without state representation.

2.4 Evaluation Protocol (P3)

To avoid subjective or text-contaminated assessments, our evaluation relies on three pillars:

1. **Structural Metrics:** Ink ratio ρ (density of non-blank glyphs), coverage (bounding box fraction), Shannon character entropy $H = -\sum p_i \log_2 p_i$, and automated flags: `spells_name` and `degenerate`.

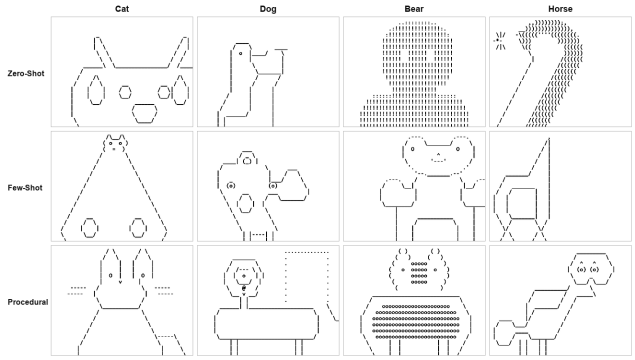


Figure 1: P1 Baseline Contact Sheet (12 Cells). Full comparison across four categories (Cat, Dog, Bear, Horse) and three prompting arms (Zero-Shot, Few-Shot, Procedural CoT). Few-shot prompting enriches stroke density; procedural prompting encourages explicit planning but exposes failure modes such as lexical leakage (e.g. spelling "HORSE" inside the canvas).

Category	Arm	ρ	Cov.	H (bits)	Chars	Spells?
Cat	zero-shot	0.056	0.49	2.60	11	No
Cat	few-shot	0.106	0.38	1.90	6	No
Cat	procedural	0.096	0.53	1.86	6	No
Dog	zero-shot	0.062	0.19	1.79	6	No
Dog	few-shot	0.094	0.42	1.89	5	No
Dog	procedural	0.097	0.38	2.41	8	No
Bear	zero-shot	0.189	0.67	1.75	6	No
Bear	few-shot	0.144	0.55	2.56	10	No
Bear	procedural	0.097	0.50	0.73	6	No
Horse	zero-shot	0.063	0.63	0.71	2	No
Horse	few-shot	0.106	0.40	1.61	5	No
Horse	procedural	0.096	0.59	2.45	12	**Yes**

Table 1: Quantitative Structural Metrics across P1 Cells. Model: Gemini 3.1 Flash-Lite. All runs parse cleanly. Procedural horse demonstrates severe lexical contamination.

2. **VLM Image-vs-Text Judge:** Dual independent queries to a frozen multimodal judge: unprompted open identification ("What does this picture depict?"), followed by a 1–5 semantic rating.
3. **Scrambled Controls & Control Gap:** For every drawing, we evaluate two controls: *scramble-cells* (uniform glyph shuffling destroying geometry) and *scramble-ink* (shuffling non-blank glyphs within the silhouette mask). The primary metric is the **Control Gap**:

$$\Delta = S_{\text{real}} - \max(S_{\text{cells}}, S_{\text{ink}}) \tag{1}$$

A positive gap $\Delta > 0$ validates genuine visual recognizability beyond lexical or density artifacts.

3 Experiments and Results

3.1 P1 Baseline Performance and Failure Modes

Table 1 reports the structural analysis across all 12 cells of the seen split. All outputs achieved 100% format contract validity.

- **Failure Mode 1: Lexical Leakage.** In the procedural `horse` cell, the model cheated by spelling the word "HORSE" across the torso (`spells_name` = True). A pure text judge falsely rewards this; our scrambled control yields $\Delta = 0.000$, correctly detecting the visual failure.
- **Failure Mode 2: Glyph Collapsing.** Zero-shot horse collapsed into 2 distinct characters ($H = 0.71$), whereas few-shot prompting encouraged rich typographic texture ($H = 1.61$–2.56).

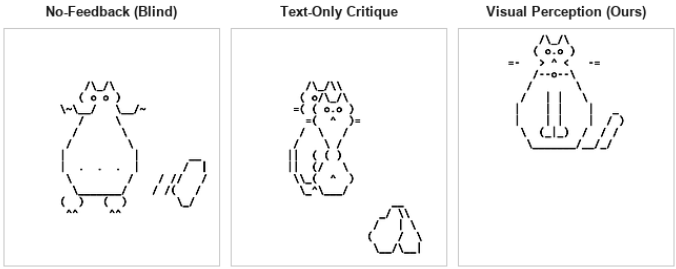


Figure 2: Feedback Mode Ablation (P2). Left: No-feedback (blind) overwrites the nose and places floating disconnected feet/tail. Center: Text-only self-critique suffers from coordinate misalignment, superimposing a second face over the first. Right: Active visual perception (ours) grounds edits to synthesize a coherent cat with whiskers, collar, paws, and tail.



Figure 3: Agentic Self-Perception Progression (P2). Turn 1: Initial silhouette draft. Turn 2: Visual critique adds symmetric whiskers and front paws. Turn 3: Final refinement adds collar and curled tail.

3.2 P2: Agentic Synthesis and Visual Grounding

As shown in Figure 3, the agentic system constructs subjects iteratively through our draft → render → perceive → revise loop. In Turn 1, the model drafts the outer head and torso silhouette. Inspecting the rendered bitmap in Turn 2, the agent identifies missing facial features and limbs, adding symmetric whiskers and seated front paws. In Turn 3, visual inspection guides the addition of a bell collar and curled tail. Without visual perception (Figure 2), multi-turn edits suffer from severe spatial drift and coordinate collisions.

3.3 P3: CLIP Evaluation and Judge Calibration

Evaluating with local CLIP ViT-B/32 against target text prompts confirms that agentic visual generation produces meaningful visual signals: `cat` achieved CLIP cosine similarity of 0.2947 vs. 0.2880 scrambled ($\Delta = +0.0067$); `bear` achieved 0.3044 vs. 0.2752 ($\Delta = +0.0293$). The dual text-vs-image judge experiment demonstrated that when evaluating purely on the rendered bitmap, the VLM judge assigned a score of 5/5 without text leak-

age.

3.4 Extensions: Editing, Photography, Animation

P4: Conversational Editing. Given a natural language request ("Add a pointy party hat with a star on top of the head"), the model issued a localized `patch_edit` targeting rows 1–4 centered at column 18. Exactly 15 cells were modified directly above the head between the ears, leaving rows 5–23 completely untouched (Figure 4a).

P5: Semantic Image-to-ASCII. Conditioning directly on a photograph, the agent extracted structural contours into a sparse 24×48 representation ($\rho = 0.143$), filtering complex textures while retaining semantic identity.

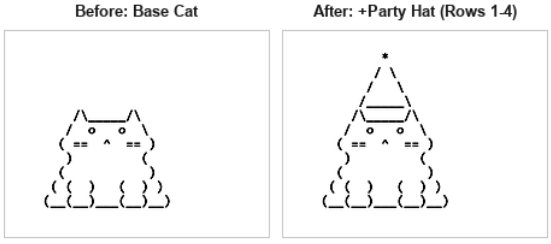
P6: Keyframe Animation. Rather than regenerating the entire grid per frame (which causes body drift), our animation loop applies localized `patch_edit` deltas onto the base keyframe. This guarantees 100% frame-to-frame invariance of the head, torso, and limbs while smoothly animating blinking eyes (`o.o` $\rightarrow$ `-.-` $\rightarrow$ `^.^`) and a wagging tail (Figure 4b).

4 Discussion and Comparisons

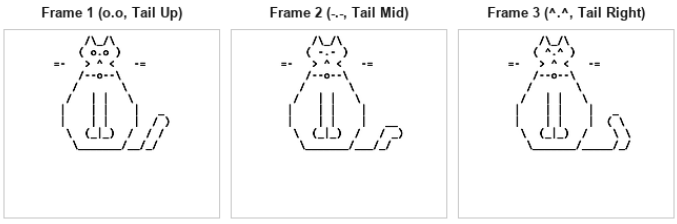
Comparison to SketchAgent and VideoSketcher. SketchAgent [4] operates in a continuous vector domain, outputting parametric Bézier coordinates stroke-by-stroke. In contrast, ASCIIgent operates over a *discrete coordinate grid* using primitive commands (`draw_rect`, `draw_line`, `patch_edit`). While SketchAgent performs single-stroke turns, ASCIIgent supports batched semantic operations followed by perceptual verification.

In VideoSketcher [5], sketch animation is addressed by having an LLM decompose a drawing plan into a sequence of strokes, which a continuous video diffusion model (e.g. SVD/AnimateDiff) animates into pen-drawing footage. In discrete ASCII art, continuous stroke diffusion does not exist. However, the core conceptual insight of VideoSketcher applies directly to our framework in two distinct ways:

1. **Drawing Process Playback:** Because ASCIIgent maintains a rigorous action history, the full drawing process can be rendered deterministically step-by-step (line by line, patch by patch), generating a discrete procedural drawing replay.
2. **Subject Motion Animation:** For semantic object motion, our P6 keyframe generator employs autoregressive visual conditioning: each frame is synthesized by feeding the previous rendered bitmap into the model, enforcing strict boundary invariance while modifying dynamic sub-regions.



(a) P4 Conversational Editing: Centered party hat added via `patch_edit` (rows 1–4) without disturbing the ears or face.



(b) P6 Delta-Patched Keyframe Animation: Blinking eyes and wagging tail with 100% invariant body and limbs.

Figure 4: Downstream Extensions. Localized editing and frame-consistent animation.

Limitations. Free-tier API quotas constrain large-scale multi-model sweeps. Additionally, while the control gap $\Delta > 0$ protects against text bias, full calibration against crowdsourced human ratings represents the gold standard.

Roadmap to CVPR. Expanding this work for a formal CVPR submission requires: (1) multi-model sweeps across proprietary and open VLMs (GPT-4o, Claude 3.5 Sonnet, LLaVA-NeXT); (2) human perceptual preference benchmarks calibrating the control gap; and (3) hybrid neuro-symbolic search over discrete edit operations.

5 Conclusion

ASCIIgent establishes a rigorous framework for studying visual abstraction in discrete text spaces. We demonstrate that monolithic generation suffers from scanline drift and lexical contamination, while agentic discrete tool execution guided by active visual perception enables grounded refinement, localized editing, and coherent keyframe animation.

References

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